Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (open to Brooklyn NY relocation)

PROFESSIONAL SUMMARY

Software engineer with 9+ years building production AI agents and data-intensive systems. Built **Leasel**, an 8-agent LangGraph HITL platform that automates complex multi-step investigation workflows with HITL escalation -- the same architecture Paces uses for due diligence automation. Built **AgentProbe**, an open-source LLM evaluation toolkit with CI regression harnesses to keep agents reliable at scale. Currently deploying Anthropic Claude (Messages API) into production financial data pipelines at Intuit. Deep Python and ETL background from 7 years at Entrata, a multi-tenant SaaS processing 180K+ federal compliance records annually. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Case studies: Leasel · AgentProbe

CORE COMPETENCIES

LangGraph Agents (Production) | Anthropic Messages API | RAG Pipelines | LLM Evaluation / Evals | Python (9+ yrs)

ETL / Data Pipelines | TypeScript / Node.js | FastAPI / REST APIs | PostgreSQL / SQL | AWS / Docker / Kubernetes

Multi-Agent Orchestration | HITL Escalation Design

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Deploy Anthropic Claude (Messages API) into production financial data pipelines -- design agent workflows, structured-output schemas, and retrieval pipelines that ground answers in source records.
- Build CI eval harnesses (LLM-as-judge + golden datasets) that block hallucinations and quality regressions before every release -- keeps agents reliable as prompts and data change.
- Build ETL pipelines with scikit-learn and XGBoost from data ingestion through feature engineering to prediction APIs; FastAPI backends and React/TypeScript front ends shipped via GitHub Actions to AWS.
- Write guardrails around LLM calls: input validation, sensitive-data handling, output checks before downstream consumption.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant SaaS, 100K+ monthly active users

- Built and owned data-intensive backend systems processing 180,000+ federal compliance records annually with validation, rollback, and reconciliation logic -- zero errors across 6 import cycles.
- Built RabbitMQ async processing for tenant rent updates across thousands of records; retry logic and idempotency eliminated double-processing in production.
- Delivered 20+ features end-to-end (Python, Java, TypeScript, React, PostgreSQL) across 120+ Agile sprints; owned requirements through deployment and production incident response.
- Ran Python/Bash automation cutting manual ops work by 80%; optimized 200+ SQL queries (30% performance gain); led Docker microservices redesign (50% better scalability).
- Owned AWS S3 PHI migration with Libsodium encryption -- 100% compliance, zero security incidents.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built data import and reconciliation workflows for 1.6M HUD/Tax Credit records; property management modules in Python, PHP, Java, PostgreSQL; led AWS S3 cloud migration.

PROJECTS

Leasel -- Multi-Agent Due Diligence Automation github.com/venkatalakshmiparimala/Lease_complaince_rag

8-agent LangGraph HITL system that automates complex investigation workflows: classifies issues, retrieves relevant rules and historical context, generates evidence-backed resolution paths, and escalates to staff when confidence is low. Hand-rolled hybrid RAG (TF-IDF + semantic reranking), cited answers, LLM-as-judge CI regression harness. The architecture directly mirrors Paces' due diligence automation: agents handle the investigation, humans approve the decisions.

Python · LangGraph (8-agent) · FastAPI · Streamlit · Hybrid RAG · SQLite · CI/CD · LLM-as-Judge · HUD API

AgentProbe -- Open-Source LLM Evaluation Toolkit github.com/venkatalakshmiparimala/AgentProbe

15-endpoint FastAPI backend routing across 5 LLM providers with LLM Router. 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance). CI regression tests on golden datasets -- the infrastructure needed to keep production agents reliable as models and prompts evolve. Runs 100% locally at zero cost.

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic · OpenAI · Groq · Ollama

AI on FHIR -- Clinical NLP & LLM API Integration github.com/venkatalakshmiparimala/AI_on_FHIR

LLM + BERT + embeddings translate English/Spanish questions into FHIR API calls against thousands of clinical data records -- reduced manual lookup time by ~60%. HIPAA-compliant role-based access control; Docker containerized.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026

GPA 3.87 · NLP pipelines, fine-tuning (HuggingFace + PyTorch), anomaly detection, regression modeling (R^2=0.977)

B.Tech Computer Science & Engineering — JNTU Anantapur 2017

TECHNICAL SKILLS

AI & Agents: LLM APIs (Anthropic Claude/Messages API, OpenAI, Groq, Ollama), LangGraph, LangChain, RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, Multi-Agent Orchestration, HITL, LLM-as-Judge Evaluation, Fine-tuning (HuggingFace, PyTorch), MCP

Languages & Web: Python (9+ yrs), TypeScript, JavaScript, Node.js, React, Java, Spring Boot, PHP, SQL, Bash

Data & Databases: PostgreSQL, MongoDB, MySQL, Redis, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, PySpark

Frameworks & APIs: FastAPI, Flask, Django, Express, REST APIs, GraphQL, RabbitMQ, Kafka, Celery

Cloud & DevOps: AWS (S3, EC2, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

ML: scikit-learn, XGBoost, PyTorch, ChromaDB, Pinecone, BERT, spaCy, HuggingFace, Prometheus, Grafana